

CHICAGO ROCKFORD, IL, USA

WMO: 725430

Lat: 42.193N

Lon: 89.093W

Elev: 730

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94822

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.2	(c) -0.5	(d) -14.2	(e) 2.6	(f) -4.7	(g) -9.2	(h) 3.4	(i) 0.2	(j) 26.0	(k) 28.1	(l) 23.8	(m) 22.8	(n) 7.2	(o) 290	(p) 0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	90.7	74.3	88.0	73.0	85.5	71.8	77.9	86.9	75.9	84.1	74.2	82.2	11.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	135.5	83.3	73.2	126.8	81.7	71.5	119.3	79.0	41.7	87.0	39.8	84.1	38.1	82.5	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	20.8	18.8	-13.5	94.9	7.2	3.2	-18.6	97.2	-22.8	99.1	-26.8	100.9	-32.1	103.2
			-14.0	80.8	6.9	2.0	-19.0	82.2	-23.1	83.4	-27.0	84.5	-32.0	86.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	21.3	25.2	37.0	49.1	60.0	69.9	73.5	71.7	64.7	52.2	38.7	27.1
	DBStd	20.45	12.36	11.90	11.26	9.17	8.48	6.62	5.59	5.12	7.70	9.07	9.78	11.17
	HDD50	3300	889	695	427	123	12	0	0	0	2	83	356	713
	HDD65	6531	1354	1113	869	482	204	28	4	7	99	408	788	1175
	CDD50	3056	0	1	25	96	323	597	728	672	442	152	18	2
	CDD65	813	0	0	2	5	50	175	266	214	89	12	0	0
	CDH74	7391	0	0	14	87	524	1639	2505	1708	792	122	0	0
	CDH80	2379	0	0	1	14	148	560	898	504	229	25	0	0
(14) Wind	WSAvg	8.8	9.7	9.9	10.3	10.9	9.4	7.8	7.0	6.5	7.3	8.8	9.6	9.3
(15) Precipitation	PrecAvg	36.6	1.5	1.3	2.5	3.9	3.9	4.8	3.8	3.9	4.1	2.8	2.6	2.0
	PrecMax	56.5	4.6	3.1	5.6	9.9	8.2	11.9	8.9	13.5	10.7	8.3	5.5	5.0
	PrecMin	23.3	0.2	0.0	0.6	1.0	0.5	0.5	0.7	0.5	0.0	0.1	0.4	0.4
	PrecStd	7.3	1.1	0.8	1.2	1.9	1.7	2.6	2.0	2.8	2.9	2.0	1.4	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	60.1	76.0	82.1	88.9	93.3	96.2	92.3	90.1	83.4	69.5	59.3
		MCWB	49.7	52.6	62.7	63.0	68.7	73.8	77.7	75.4	72.9	68.2	58.2	56.7
	2%	DB	45.4	50.5	66.3	76.0	84.2	89.4	90.9	88.4	86.2	77.2	62.8	51.9
		MCWB	41.7	45.0	56.6	60.4	67.4	73.2	76.1	73.6	71.4	63.5	54.9	48.5
	5%	DB	40.2	44.7	59.9	70.5	80.1	86.4	87.8	85.3	82.2	72.0	58.2	45.9
		MCWB	37.6	40.3	52.2	56.7	65.4	71.5	74.2	72.7	68.7	61.4	52.6	42.6
	10%	DB	36.7	39.9	53.7	65.6	75.7	83.0	85.1	82.5	78.7	67.3	53.7	41.3
		MCWB	34.5	36.7	46.6	54.2	63.3	69.7	72.6	70.8	66.5	59.0	48.1	38.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.9	53.9	63.7	65.8	73.1	78.0	81.3	78.7	76.5	70.5	60.9	57.1
		MCDB	52.6	57.4	74.4	76.2	83.5	87.7	89.9	88.1	85.8	79.3	65.9	59.0
	2%	WB	42.1	45.8	58.6	62.1	70.2	75.4	78.7	76.7	73.4	66.6	57.0	48.8
		MCDB	44.9	50.0	65.2	72.5	79.8	85.0	87.4	84.7	81.3	73.1	61.1	50.9
	5%	WB	37.8	40.7	53.2	59.1	67.8	73.5	76.8	74.9	71.1	63.4	52.8	43.1
		MCDB	40.1	44.2	58.6	68.5	76.6	83.4	85.3	81.8	79.1	70.0	57.2	45.4
	10%	WB	34.6	37.0	46.9	55.2	65.2	71.6	74.7	72.9	68.9	60.0	48.5	38.6
		MCDB	36.6	39.8	53.1	64.0	73.5	80.1	82.5	79.7	76.0	66.4	53.3	41.2
(35) Mean Daily Temperature Range	MDBR		14.2	15.3	17.7	20.6	20.7	20.0	19.5	19.1	21.2	19.6	16.1	13.5
	5% DB	MCDBR	17.4	19.8	25.0	28.5	25.8	23.3	21.6	21.4	24.3	25.6	22.7	18.0
		MCWBR	14.7	15.2	16.5	16.4	12.8	10.9	10.1	10.1	11.6	14.2	16.2	14.9
	5% WB	MCDBR	16.4	18.8	23.1	25.0	21.7	20.3	19.6	18.4	20.6	21.3	20.8	16.4
		MCWBR	14.4	15.0	16.7	16.4	12.5	10.9	10.9	10.4	11.1	14.7	16.5	14.7
(40) Clear-Sky Solar Irradiance	taub		0.285	0.304	0.334	0.386	0.414	0.428	0.440	0.419	0.391	0.341	0.301	0.282
	taud		2.397	2.359	2.396	2.263	2.237	2.244	2.202	2.258	2.313	2.454	2.505	2.439
	Ebn at Noon		274	285	289	280	273	269	264	266	266	269	267	265
	Edh at Noon		25	30	33	41	43	43	45	41	36	28	23	22
(44) All-Sky Solar Radiation	RadAvg		539	846	1168	1473	1709	1937	1933	1700	1410	920	596	438
	RadStd		67	102	109	121	120	142	108	104	117	110	51	42

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+86	
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page